

FORM FACTOR EVOLUTION AND FUTURE WEARABLE EMBODIMENTS

General Form Factor Flexibility

The longitudinal penile physiology monitoring system may be implemented in a wide variety of physical form factors beyond any single ring configuration. The disclosed invention contemplates that physiologic sensing functionality may be embodied in structures optimized for comfort, manufacturability, miniaturization, user preference, or evolving wearable design paradigms.

While certain embodiments may utilize closed-loop rings or open-gap wearable structures, alternative embodiments may include flexible bands, adjustable loops, segmented structures, expandable frames, soft wearable sleeves, textile-integrated devices, adhesive-mounted sensors, or conformal wearable systems capable of maintaining physiologic contact with the monitored region.

Adaptive and Dynamic Structures

Future embodiments may incorporate adaptive or dynamically adjustable structures capable of modifying shape, stiffness, or contact pressure in response to physiologic changes. Such structures may include inflatable elements, mechanically actuated frames, shape-memory materials, soft robotic components, or tension-adjusting mechanisms designed to maintain optimal sensing conditions.

Adaptive structures may improve measurement stability, comfort during extended wear, and accuracy across varying physiologic states without requiring manual repositioning by the user.

Miniaturization and Distributed Wearables

The invention further contemplates miniaturized embodiments in which sensing components are distributed across multiple small wearable elements rather than a single device body. Distributed wearables may communicate wirelessly to function as a coordinated sensing network capable of collectively performing longitudinal physiologic monitoring.

Miniaturized implementations may leverage flexible electronics, printed sensor substrates, microelectromechanical systems, or future ultra-small sensing technologies enabling reduced device visibility and enhanced user comfort.

Integration with Apparel and Accessories

Embodiments may integrate sensing functionality into apparel, undergarments, support garments, medical wearables, or accessory components positioned near the monitored anatomy. Integration may allow physiologic monitoring without requiring a dedicated standalone wearable structure.

Future Wearable Technologies

The disclosure expressly includes wearable technologies not yet developed at the time of filing, including emerging flexible electronics, bio-integrated sensors, energy-harvesting wearables, and novel human-interface materials capable of supporting longitudinal physiologic monitoring.

The form factor embodiments described herein are illustrative and non-limiting and encompass any physical implementation capable of supporting multi-modal sensing of longitudinal penile physiology.